Vikas Pal

Forward Deployed Engineer | Applied AI × Fintech · Platform, real-time trading infrastructure, production LLM systems

Ahmedabad, India (IST) · open to remote, hybrid or on-site worldwide · palv499@gmail.com · linkedin.com/in/vikaspal1704 · github.com/vikaspal1704 · cal.com/vikas-pal-2mz8du · vikaspal.is-a.dev

SUMMARY

Software engineer with 4+ years shipping production systems where fintech meets AI. I created, and lead, a brokerage-infrastructure company's unified front-end platform, which merged 7 frontends into one and serves 300+ clients. Previously ran an LLM platform at 99.9% uptime while cutting model costs ~40%. In open source: a Rust matching engine at ~2M orders/s.

EXPERIENCE

ViewTrade Holding Corp. · Software Development Engineer I Aug 2025 – Present

GIFT City, Gandhinagar · API-driven brokerage and wealth-tech infrastructure

- Conceived and built, from scratch, the company's **unified, configurable front-end platform**, and lead it with a **global team** of US engineers and external vendors.
- **Merged 7 frontend codebases into one platform serving 300+ advisory and retail clients**, and brought internal frontends onto the same architecture. Clients configure layouts for their firm and users without new code.
- Build real-time trading paths over **WebSockets and distributed systems** on **.NET and AWS**, handling stale quotes, fan-out and backpressure where correctness and uptime come first.
- Drive adoption of AI-native, Claude-ecosystem tooling for developer productivity without giving up reliability.

Zeus Learning · Software Developer Jan 2023 – Jul 2025

Mumbai · AI-powered test platform (Next.js, Python/Django, .NET, AWS)

- Built and ran the platform at **~50,000 requests/day with 99.9% uptime** under peak load.
- Engineered **semantic caching that cut LLM costs ~40%** without changing the model family.
- **Led 4 developers** to deliver a reporting SDK under hard client deadlines.
- Set up CI/CD with DevOps that made **deploys ~50% faster**; modular patterns cut debugging time ~20%.

CultureX Entertainment · Product Development Engineer Jul 2022 – Dec 2022

- Built web pages and product features for CreatorX in a team of 4; code reviews and pre-release bug fixing.

SELECTED OPEN-SOURCE PROJECTS

- **Arena**: a stock exchange in the browser. Dependency-free Rust matching engine compiled to an 80 KB WebAssembly module; **~2M orders/s, p50 ≈ 350 ns, p99 ≈ 1.6 µs** on one core. Differential-tested in CI against an independent Python engine on 1.5M random commands. Journal replay with a fingerprint chain; profiling-led fix took p99 from 16.8 µs to 1.6 µs.
- **FIX Protocol Lab**: FIX codec and TCP session engine from scratch (FIX 4.2–5.0 SP2) with resend and gap fill, plus a live visual front end; Node.js, TypeScript, React, Playwright.
- **F&O Wrapped**: privacy-first P&L analytics for Indian F&O traders across 4 brokers; FIFO in integer paise, fully in-browser PWA.
- **Recruiter Backend**: FastAPI service for LLM resume parsing and Pinecone semantic search; auth, mocked tests, Docker CI.
- **Live Orderbook Feed** (WebSocket L2 fan-out with backpressure) and **this portfolio** (on-device LLM via WebLLM, retrieval, JD fit check).

SKILLS

Languages	Python, C# / .NET, TypeScript / JavaScript, Rust, SQL
Backend	.NET, Django, FastAPI, Node.js, distributed systems, microservices, WebSockets, TCP, PostgreSQL
AWS	EC2, ECS / EKS, Lambda, API Gateway, S3, RDS, DynamoDB, SQS, SNS, Kinesis
Frontend	Front-end platform architecture, React, Next.js, Redux Toolkit, WebAssembly, Web Workers
AI / LLM	Production LLM platforms, semantic caching, RAG and vector search (Pinecone), WebLLM / WebGPU, Claude tooling
Fintech	Matching engines, L2 market data, FIX, P&L and charges, brokerage / wealth-tech APIs
Quality	Docker, GitHub Actions CI/CD, pytest, Vitest, Playwright, differential and golden-vector testing

EDUCATION

BE in Computer Science · LDRP Institute of Technology & Research (Gujarat Technological University) 2019 – 2023